Problem-set EE322M

Q 1: Find the inverse Fourier Transform of

$$X(\omega) = \frac{j\omega}{(2+j\omega)^2}$$

Solution: Note that

$$\mathcal{F}(e^{-2t}u(t)) = \frac{1}{2+i\omega}$$

Using Frequency Differentiation property-

$$\mathcal{F}\left(-jte^{-2t}u(t)\right) = \frac{d}{d\omega}\left(\frac{1}{2+i\omega}\right) = \frac{1}{(2+i\omega)^2}$$

Now, by using time differentiation property,

$$\mathcal{F}\left(\frac{d}{dt}te^{-2t}u(t)\right) = \frac{j\omega}{(2+i\omega)^2}$$

Therefore,

$$x(t) = \frac{d}{dt}te^{-2t}u(t) = (1-2t)e^{-2t}u(t)$$

Q 2: Given an input sequence

$$x[n] = 6\delta(n+2) + 3\delta(n+1) + 8\delta(n) + 7\delta(n-1) + 4\delta(n-2)$$

Find $\frac{1}{\pi}\int\limits_{-\pi}^{\pi} X(e^{j\omega})\sin^2(2w)d\omega$, where $X(e^{j\omega})$ is the frequency response of $x(t)$.

Solution:

$$\frac{1}{\pi}\int\limits_{-\pi}^{\pi} X(e^{j\omega})\sin^2(2w)d\omega = \frac{1}{\pi}\int\limits_{-\pi}^{\pi} X(e^{j\omega})\frac{(1-\cos(4w))}{2}d\omega$$

Simplifying using $\cos(\theta) = \frac{e^{j\theta}+e^{-j\theta}}{2}$, we have

$$\begin{aligned}\frac{1}{\pi}\int\limits_{-\pi}^{\pi} X(e^{j\omega})\sin^2(2w)d\omega &= \frac{1}{2\pi}\int\limits_{-\pi}^{\pi} X(e^{j\omega})d\omega - \frac{1}{2}\left[\frac{1}{2\pi}\int\limits_{-\infty}^{\infty} X(e^{j\omega})e^{(4j\omega)}d\omega + \frac{1}{2\pi}\int\limits_{-\infty}^{\infty} X(e^{j\omega})e^{(-4j\omega)}d\omega\right] \\ &= x(0) - \frac{1}{2}[x(4) - x(-4)] = 8.\end{aligned}$$

Q 3: Find minimum sampling rate of y(t) given in Figure 1.

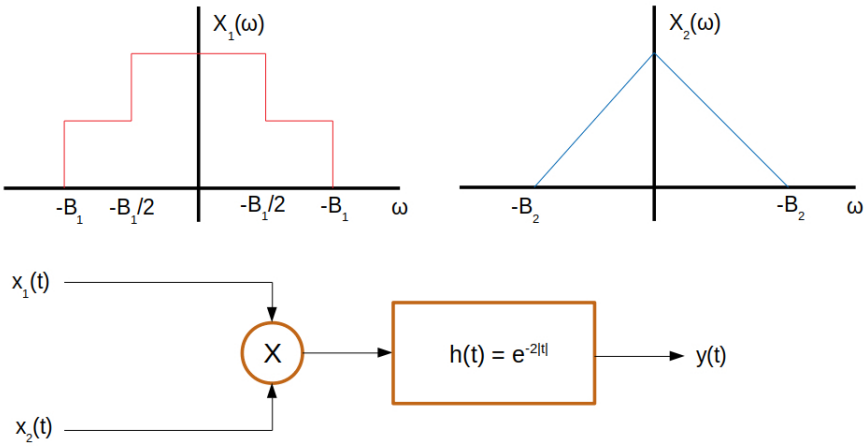

Figure 1: Question 3

Solution: $x(t) = x_1(t) + x_2(t) \Rightarrow X(\omega) = X_1(\omega) + X_2(\omega)$. The frequency of $X(\omega)$ will be in the range $[-(B_1 + B_2), (B_1 + B_2)]$. Further,

$$H(w) = \left(\frac{4}{w^2 + 4}\right) \text{ with } \omega \text{ in the range } (-\infty, \infty).$$

Since $Y(\omega) = X(\omega)H(\omega)$, the maximum frequency of $Y(\omega) = B_1 + B_2$. Hence, minimum sampling rate $= 2(B_1 + B_2)$.

Q 4: Given an input sequence $x[n] = \delta(n+1) + 2\delta(n) + \delta(n-1)$. Find

$$\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega,$$

where $X(j\omega)$ is the Fourier transform of $x(t)$.

Solution: Using Parseval's Theorem: $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = 2\pi(1^2 + 2^2 + 1^2) = 12\pi$.

Q 5: For a signal $x(t)$ that is time-limited to 10 ms and has a frequency range from 0 kHz to 10 kHz, determine N, the number of signal samples necessary to compute a radix-2 FFT with a frequency resolution f_0 of at least 50 Hz. Explain whether any zero padding is necessary.

Solution: The frequency resolution is the difference in frequency between each sample, and thus sets a limit on how precise the results can be. The frequency resolution is equal to the sampling frequency divided by FFT size.

Since maximum frequency $= 10$ kHz. We have $f_s \geq 20$ kHz. Let us choose $f_s = 20$ kHz. Further we want a frequency resolution of atleast 50 Hz. Therefore,

$$N = \frac{20 \text{ kHz}}{50 \text{ Hz}} = 400.$$

Since N must be a power of 2, the minimum value of N has to be 512 to implement the radix-2 FFT algorithm.

Further, the sample distance is $T = \frac{1}{f_s} = \frac{1}{20000} = 50\mu s$. Therefore, the total duration of $x(t)$ needs to be $512 \times 50\mu s = 25.6$ ms. Since $x(t)$ is of 10 ms duration, we need zero padding over 15.6 ms. With this configuration the frequency resolution obtained is $\frac{20000}{512} = 39.0625$ Hz.

Alternatively, we could also have changed the sampling frequency, keeping $N = 512$. In that case

$$f_s = 50 \times 512 = 25600 \text{ Hz}.$$

The sample distance in that case is $T = \frac{1}{f_s} = \frac{1}{25600} = 39.0625\mu s$. Therefore, the total duration of $x(t)$ needs to be $512 \times 39.0625\mu s = 20$ ms. Since $x(t)$ is of 10 ms duration, we need zero padding over 10 ms.

There are also other possibilities of increasing the frequency resolution.

Q 6: A designer has available a number of eight-point FFT chips. Show explicitly how he should interconnect three such chips in order to compute a 24-point DFT.

Solution: Create three subsequences of 8-points each

$$\begin{aligned}
Y[k] = \sum_{n=0}^{2} 3y[n]W_N^{kn} &= \sum_{n=0,3,6,\ldots}^{21} y[n]W_N^{kn} + \sum_{n=1,4,7,\ldots}^{22} y[n]W_N^{kn} \sum_{n=2,5,\ldots}^{23} y[n]W_N^{kn} \\
&= \sum_{\ell=0}^{7} y[3\ell]W_{\frac{N}{3}}^{k\ell} + \sum_{\ell=0}^{7} y(3\ell+1)W_{\frac{N}{3}}^{k\ell}W_N^{k} + \sum_{\ell=0}^{7} y(3\ell+2)W_{\frac{N}{3}}^{k\ell}W_N^{2k} \\
&= \underbrace{\sum_{\ell=0}^{7} y[3\ell]W_{\frac{N}{3}}^{k\ell}}_{Y_1[k]} + W_N^{k}\underbrace{\sum_{\ell=0}^{7} y(3\ell+1)W_{\frac{N}{3}}^{k\ell}}_{Y_2[k]} + W_N^{2k}\underbrace{\sum_{\ell=0}^{7} y(3\ell+2)W_{\frac{N}{3}}^{k\ell}}_{Y_3[k]},
\end{aligned}$$

where $Y_1[k], Y_2[k], Y_3[k]$ represent the 8-point DFTs of the subsequences. Draw the interconnection on your own.

Q 7:

(a) Suppose that $x[n]$ is a finite-duration sequence of $N = 1024$ points. It is desired to evaluate the $z-$transform $X(z)$ of the sequence at the points

$$z_k = e^{j(\frac{2\pi}{1024})k}, k = 0, 100, 200, ..., 1000.$$

by using the most efficient method or algorithm possible. Describe an algorithm for performing this computation. Compute the number of operations involved in the algorithm.

(b) Repeat part (a) if $X(z)$ is to be evaluated at

$$z_k = 2 \times (0.9)^k \times e^{j[(\frac{2\pi}{5000})k + \frac{\pi}{2}]}, k = 0, 1, 2, ..., 999.$$

Solution: FFT will work. How you figure out.

Q 8: Consider the signal

$$x[n] = a^n u[n] \qquad 0 < a < 1. \tag{1}$$

The spectrum of this signal is sampled at frequencies $w_k = 2\pi k/N, \quad k = 0, 1, ..., N-1$. Determine the reconstructed spectra for $a = 0.8$ when $N = 5$ and $N = 20$.

Solution: The Fourier transform of the sequence $x(n)$ is

$$X(w) = \sum_{n=0}^{\infty} a^n e^{-jwn} = \frac{1}{1 - ae^{-jw}} \tag{2}$$

Suppose then we sample $X(w)$ at N equidistant frequencies $w_k = 2\pi k/N, k = 0, 1, ..., N-1$. Thus we obtain the spectral samples

$$X[k] = X\left(\frac{2\pi k}{N}\right) = \frac{1}{1 - ae^{-j2\pi k/N}}, \qquad k = 0, 1, ..., N-1. \tag{3}$$

The periodic sequence $x_p(n)$, corresponding to the frequency samples $X[k]$, $k = 0, 1, ..., N-1$ can be obtained from

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n - lN) = \sum_{l=-\infty}^{0} a^{n-lN} \tag{4}$$

$$= a^n \sum_{l=0}^{\infty} a^{lN} = \frac{a^n}{1 - a^N} \qquad 0 \le n \le N-1 \tag{5}$$

Where the factor $\frac{a^n}{1-a^N}$ represents the effect of aliasing. Since $0 < a < 1$, the aliasing error tends towards zero as $N \to \infty$. See the figure 14.

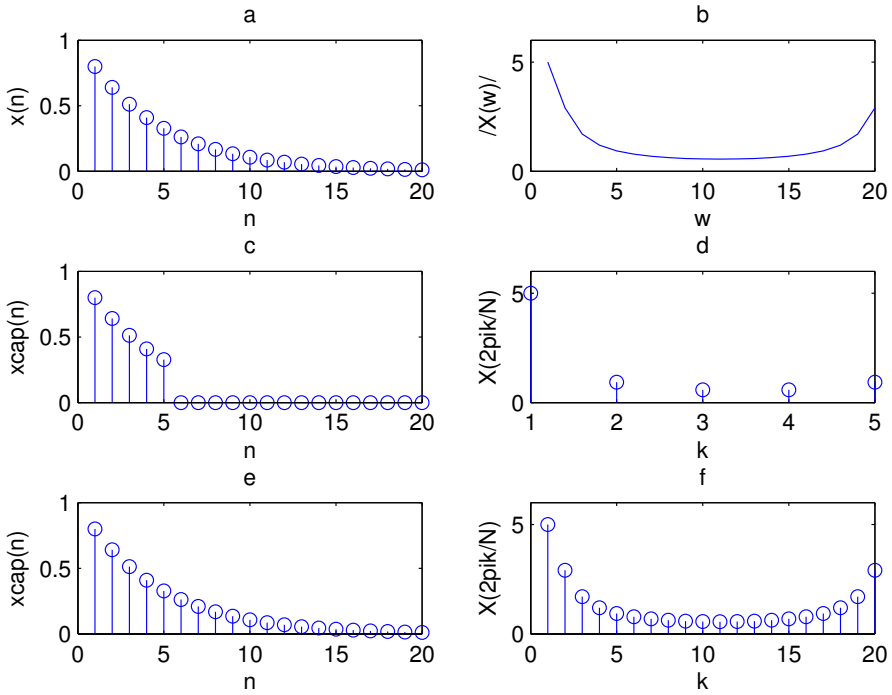

Figure 2: (a) Plot of sequence $x(n) = (0.8)^n u(n)$; (b) Its Fourier transform (magnitude only); (c) and (d) effect of aliasing with $N = 5$; (e) and (f) reduced effect of aliasing with $N = 20$

If we define the aliased finite-duration sequence $X(n)$ as

$$\hat{x}(n) = \begin{cases} x_p(n), & 0 \le n \le N-1 \\ 0, & otherwise \end{cases} \tag{6}$$

then its Fourier transform is

$$\hat{X}(w) = \sum_{n=0}^{N-1} \hat{x}(n)e^{-jwn} = \sum_{n=0}^{N-1} x_p(n)e^{-jwn} \tag{7}$$

$$= \frac{1}{1-a^N} \cdot \frac{1-a^n e^{-jwN}}{1-ae^{-jw}} \tag{8}$$

Note that although $\hat{X}(w) \neq X(w)$, the sample values at $w_k = 2\pi k/N$ are identical. That is

$$\hat{X}(2\pi k/N) = \frac{1}{1-a^N} \cdot \frac{1-a^N}{1-ae^{-j2\pi k/N}} = X\left(\frac{2\pi k}{N}\right). \tag{9}$$

Q 9: Let us consider a signal $x[n]$, derived by sampling a signal $x(t)$, at a sampling frequency F_s. Let $x(t)$ have a finite support from 0 to T.

(i) Using Discrete Time Fourier Series (DTFS), obtain the Discrete Fourier Transform (DFT) of $x[n]$.

(ii) What is the time and frequency resolution of the signal $x[n]$?

Solution: (i) This is already done in one of the modules.
(ii) The time resolution of the signal is defined as the interval over which its Fourier Transform is defined. Figure out the answer.

The frequency resolution of the signal $x[n]$ is defined as the frequency spacing of two adjacent frequencies in its DFT spectrum. Figure out the answer.

Q 10: Compute the DTFT of the following $x[n] = 2^{|n|}$.

Solution: Does not exist. Figure out why.

Q 11: The butterfly in Figure 3 was taken from a decimation-in-frequency FFT with $N = 16$, where the input sequence was arranged in normal order. Note that a 16-point FFT will have four stages, indexed $m = 1, ..., 4$. Which of the four stages have butterflies of this form? Justify your answer.

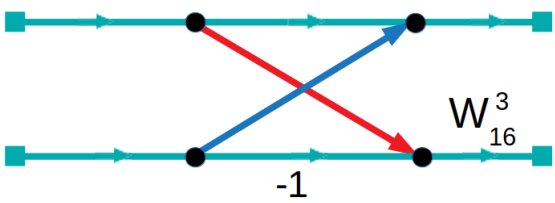

Figure 3: Question 11

Solution: Only the $m = 1$ stage will have this form. No other stage of a $N = 16$ radix-2 decimation-in-frequency FFT will have a W_{16} term raised to an odd power.

Q 12: Suppose you have two programs for computing the DFT of a sequence $x[n]$ that has $N = 2^\beta$ nonzero samples. Program A computes the DFT by directly implementing the definition of the DFT sum. Program B implements the decimation-in-time FFT algorithm. What is the shortest sequence N such that Program B runs faster than Program A?

Solution: $N = 64$.

Q 13: Suppose you have a signal $x[n]$ with 1021 nonzero samples whose discrete-time Fourier transform you wish to estimate by computing the DFT. You find that it takes your computer 100 seconds to compute the 1021-point DFT of $x[n]$. You then add three zero-valued samples at the end of the sequence to form a 1024-point sequence $x_1[n]$. The same program on your computer requires only 1 second to compute $X_1[k]$. Reflecting, you realize that by using $x_1[n]$, you are able to compute more samples of $X(e^{j\omega})$ in a much shorter time by adding some zeros to the end of $x[n]$ and pretending that the sequence is longer. How do you explain this apparent paradox?

Solution: Hint: Use of FFT.

Q 14: The N-point DFT of the N-point sequence $x[n] = e^{-j(\pi/N)n^2}$, assuming that N is even, is

$$X[k] = \sqrt{N} e^{-j\pi/4} e^{j(\pi/N)k^2}.$$

Determine the $2N$-point DFT of the $2N$-point sequence $y[n] = e^{-j(\pi/N)n^2}$, assuming N is even.

Solution: Use the DFT formula for $Y[k]$ to note that

$$Y[k] = \begin{cases} 2X[k/2], & k \text{ even}, \\ 0, & k \text{ odd}. \end{cases}$$

Q 15: In Module-12c, we showed that linear time-invariant filtering corresponding to long sequences can be done using overlap-add and the overlap-save methods. If the DFTs in these method are computed using an FFT algorithm, these sectioning methods can require fewer complex multiplications per output sample than the direct evaluation of the convolution sum.

(i) Assume that the complex input sequence $x[n]$ is of infinite duration and that the complex impulse response $h[n]$ is of length P samples, so that $h[n] \neq O$ only for $O \leqslant n \leqslant P-1$. Also, assume that the output is computed using the overlap-save method, with the DFTs of length $L = 2^\beta$, and suppose that these DFTs are computed using a radix-2 FFT algorithm. Determine an expression for the number of complex multiplications required per output sample as a function of β and P.

(ii) Suppose that the length of the impulse response is $P = 500$. By evaluating the formula obtained in part (i), plot the number of multiplications per output sample as a function of β for the values of $\beta \leqslant 20$ such that the overlap-save method applies. For what value of β is the number of multiplications minimal? Compare the number of complex multiplications per output sample for the overlap-save method using the FFT with the number of complex multiplications per output sample required for direct evaluation of the convolution sum.

(iii) Show that for large FFT lengths, the number of complex multiplications per output sample is approximately β. Thus, beyond a certain FFT length, the overlap-save method is less efficient than the direct method. If $P = 500$, for what value of β will the direct method be more efficient?

(iv) Assume that the FFT length is twice the length of the impulse response, i.e., $L = 2P$, and assume that $L = 2^\beta$. Using the formula obtained in part (i), determine the smallest value of P such that the overlap-save method using the FFT requires fewer complex multiplications than the direct convolution method.

Solution:

(i) $\dfrac{\text{Complex multiplications}}{\text{Output sample}} = \frac{2^\beta(\beta+1)}{2^\beta - P + 1}$ (assuming $H[k]$ is pre-calculated).

(ii) $\beta = 9$.

(iii) Take $\beta \to \infty$ in the expression for $\dfrac{\text{Complex multiplications}}{\text{Output sample}}$. For $P = 500$, direct method will be more efficient for $\beta > 500$.

(iv) $\beta = 5$ or $P = 16$.

Q 16: Suppose $x_c(t)$ is a periodic continuous-time signal with period 1 ms and for which the Fourier series is

$$x_c(t) = \sum_{k=-9}^{9} a_k e^{j(2\pi k t/10^{-3})}$$

The Fourier series coefficients a_k are zero for $|k| > 9$. $x_c(t)$ is sampled with a sample spacing $T = \frac{1}{6} \times 10^{-3}$ sec to form $x[n]$. That is

$$x[n] = x_c\left(\frac{10^{-3} n}{6}\right).$$

(a) Is $x[n]$ periodic and, if so, with what period?

(b) Is the sampling rate above the Nyquist rate?

(c) Find the discrete Fourier series coefficients of $x[n]$ in terms of a_k.

Solution:

(a) The sequence $x[n]$ is periodic with period 6.

(b) The Nyquist Criterion is violated.

(c) The coefficients are $6 \sum_{r=-\infty}^{\infty} a_{k-6r}$.

Q 17: Consider a 20-point finite-duration sequence $x[n]$ such that $x[n] = 0$ outside $0 \leqslant n \leqslant 19$, and let $X(e^{j\omega})$ represent the Fourier transform of $x[n]$.

(a) If it is desired to evaluate $X(e^{j\omega})$ at $\omega = 4\pi/5$ by computing one M-point DFT, determine the smallest possible M, and develop a method to obtain $X(e^{j\omega})$ at $\omega = 4\pi/5$ using the smallest M.

(b) If it is desired to evaluate $X(e^{j\omega})$ at $\omega = 10\pi/27$ by computing one L-point DFT, determine the smallest possible L, and develop a method to obtain $X(e^{j10\pi/21})$ using the smallest L.

Solution:

(a) Since we want to evaluate $X(e^{j\omega})$ at $\omega = 4\pi/5$, hence a possible size of the DFT is evident by the periodicity of $e^{j\omega}|_{\omega=4\pi/5}$. Suppose we choose $M = 5$. The desired value occurs at $k = 2$.

(b) The smallest DFT size is $L = 27$. Since DFT is larger than the data block, $x[n]$ needs to be padded with zeros. The desired value occurs at $k = 5$.

Q 18: The two eight-point sequences $x_1[n]$ and $x_2[n]$ are as shown in Figure 4 have DFTs $X_1[k]$ and $X_2[k]$, respectively.

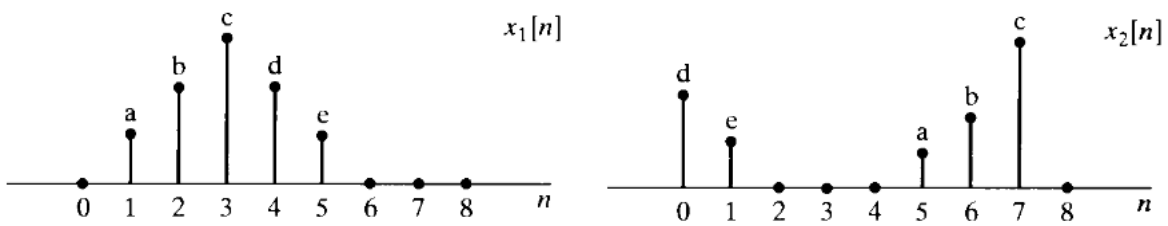

Figure 4: Question 18

Determine the relationship between $X_[k]$ and $X_2[k]$.

Solution: $X_2[k] = (-1)^k X_1[k]$.

Q 19: Figure 5 shows two sequences $x_1[n]$ and $x_2[n]$. The value of $x_2[n]$ at time $n = 3$ is given by a. Figure 6 shows $y[n]$, the four point circular convolution of $x_1[n]$ and $x_2[n]$. What is the value of a?

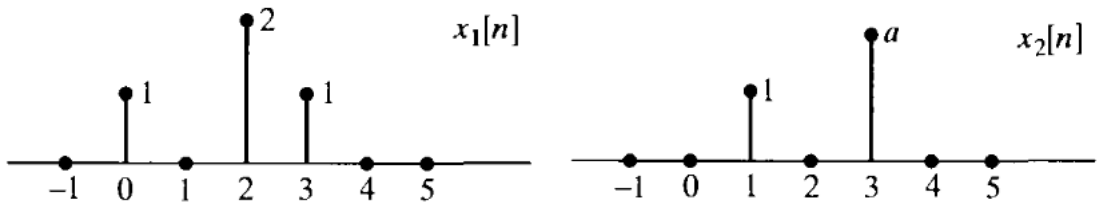

Figure 5: Question 19

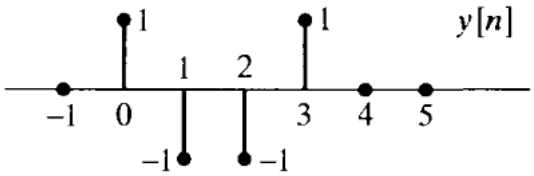

Figure 6: Question 19

Solution: $a = -1$.

Q 20: Consider an LTI system with $|H(e^{j\omega})| = 1$, and let $\texttt{arg}[H(e^{j\omega})]$ be as shown in Figure 7. If the input is

$$x[n] = \cos\left(\frac{3\pi}{2}n + \frac{\pi}{4}\right),$$

determine the output $y[n]$.

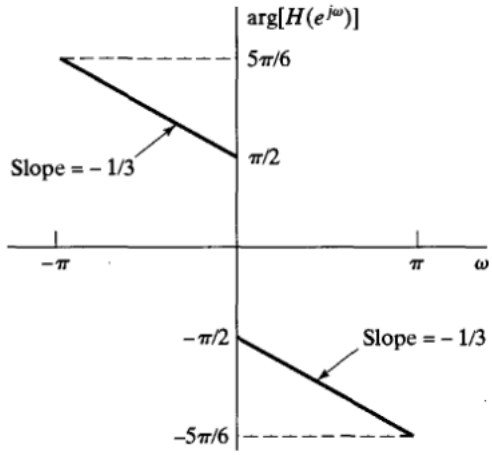

Figure 7: Question 20

Solution: $y[n] = \cos\left(\frac{3\pi}{2}n + \frac{11\pi}{12}\right)$.

Q 21: For the system in Figure 8, determine the output $y[n]$ when the input $x[n]$ is $\delta[n]$ and $H(e^{j\omega})$ is an ideal low pass filter as indicated, i.e.,

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \pi/2, \\ 0, & \pi/2 < |\omega| \leqslant \pi. \end{cases}$$

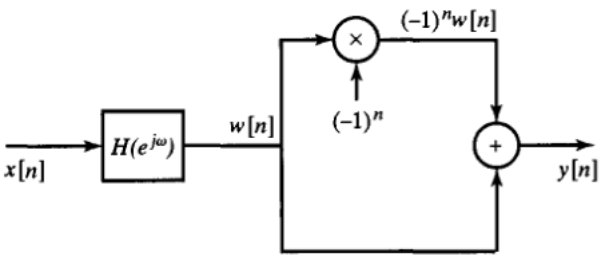

Figure 8: Question 21

Solution: $y[n] = \delta[n]$.

Q 22: Consider an LTI system with frequency response

$$H(e^{j\omega}) = e^{-j[(\omega/2)+(\pi/4)]}, \qquad -\pi < \omega \geqslant \pi.$$

Determine $y[n]$, the output of this system, if the input is

$$x[n] = \cos\left(\frac{15\pi n}{4} - \frac{\pi}{3}\right)$$

for all n.

Solution: $y[n] = e^{-j\pi/4} \cos\left(\frac{\pi n}{4} + \frac{5\pi}{24}\right)$.

Q 23: Consider an LTI system whose frequency response is

$$H(e^{j\omega}) = e^{-j\omega/2}, \qquad |\omega| < \pi.$$

Determine whether or not the system is causal.

Solution: Let the input be $x[n] = \delta[n-1]$, if the system is causal then the output, $y[n]$, should be zero for $n < 1$. So we evaluate $y[0]$ and check. The system is not causal.

Q 24: We would like to linearly convolve a 3000-point sequence with a linear shift-invariant filter whose unit sample response is 60 points long. To utilize the computational efficiency of the fast Fourier transform algorithm, the filter is to be implemented using 128-point discrete Fourier transforms and inverse discrete Fourier transforms. If the overlap-add method is used, how many DFTs are needed to complete the filtering operation?

Solution: Here length of $h[n]$ is $L60$ and $N = 128$. Therefore, $x[n]$ must be partitioned into sequences of length $M = N - L + 1 = 69$. Because $x[n]$ is 3000 points long, we will have 44 sequences (with the last sequence containing only 33 nonzero values). Thus, to perform the convolution we need:

1. One DFT to compute $H[k]$.
2. 44 DFTs for $X_i[k]$.
3. 44 inverse DFTs for $Y_i[k] = H[k]X_i[k]$.

A total of 45 DFTs and 44 inverse DFTs are required.

Q 25: Suppose that we are given 10 s of speech that has been sampled at a rate of 8 kHz and that we would like to filter it with an FIR filter $h[n]$ of length $L = 64$. Using the overlap-save method with 1024-point DFTs, how many DFTs and inverse DFTs are necessary to perform the convolution?

Solution: 85 DFTs and 84 inverse DFTs.